ReSCD-Net: Resolution-Aware Deep Learning for Building Change Detection in High-Resolution Satellite Imagery

Mohamed Mosaad Elisy, Xiongwu Xiao, *Member, IEEE*, Mohamed H. Sharouda, Jianya Gong, *Member, IEEE*, and Deren Li

***Abstract*—Building change detection (BCD) using high-resolution satellite data is critical in urban surveillance, disaster management, and environmental assessments. However, issues such as resolution inconsistencies, spectrum fluctuation, and urban heterogeneity frequently restrict the reliability of current approaches. We proposed a novel resolution-standardized change detection network (ReSCD-Net) to address these problems. The framework features an adaptive resolution normalizer (ARN) for dynamic resolution standardization, a resolution-aware segmentation core (RASC) with a hierarchical spatial extractor (HSE) for semantic representation at various scales, and a contextual refinement decoder (CRD) for precise boundary reconstruction. Multiscale characteristics are further enhanced by atrous spatial pyramid pooling (ASPP), allowing for robust contextual reasoning and detailed edge preservation. In addition, a balanced structural loss (BSL) function is added to improve the tradeoff between precision and recall while suppressing spurious changes caused by shadows, seasonal variations, and sensor noise. Three benchmark datasets covering various architectural and geographic circumstances [Egypt BCD dataset (EGY-BCD), LEVIR-CD+, and deeply supervised image fusion network detection dataset (DSIFN-CD)] were used for the experiments. Achieving an $F1$-score of 91.09% and a mean intersection over union (mIoU) of 91.12% on the EGY-BCD dataset, ReSCD-Net outperformed recent convolutional neural network (CNN)- and Transformer-based baselines. With an $F1$-score of 93.06% and an mIoU of 90.83% on LEVIR-CD+, the model surpassed transformer-enhanced techniques while preserving a lower computing complexity. ReSCD-Net's balanced precision and recall scores on DSIFN-CD further validated its capacity for generalization in diverse urban and rural settings.**

***Index Terms*—Building change detection (BCD), deep learning, multitemporal image analysis, semantic segmentation.**

I. Introduction

HIGH-RESOLUTION satellite imagery has elevated building change detection (BCD) to the leading edge of remote sensing research, with applications in infrastructure management, urban planning, environmental monitoring, and disaster response. Rapid urbanization, population increase, and anthropogenic activities necessitate precise monitoring of constructed areas in order to plan for long-term development and resilience. BCD, defined as the detection of structural changes over time, has made tremendous progress with the availability of high-resolution datasets and deep learning algorithms [1].

BCD aids in urban growth monitoring, land-use mapping, and accurate geospatial databases, enabling informed zoning, infrastructure development, and water management through impervious surface analysis [2], [3]. Beyond urban areas, it assists in environmental change monitoring, disaster damage assessment, and agricultural planning [4], [5]. Very-high-resolution (VHR) optical data from satellites such as WorldView and QuickBird enable fine-scale building studies; however, accuracy is hampered by radiometric changes, seasonal impacts, shadows, and misregistration errors [6], [7].

Traditional pixel-based approaches, such as image differencing, principal component analysis (PCA), and postclassification comparison [8], [9], provided initial findings but were limited by noise and adaptability in complex urban contexts. Object-based methods and traditional machine learning enhanced spatial context integration [10], [11] but struggled with hierarchical feature learning. The use of deep learning, including convolutional neural networks (CNNs) such as U-Net and its derivatives, has resulted in significant improvements in spatial feature extraction. Siamese networks advanced bitemporal modeling with shared encoders and similarity learning [12], [13], while transformer-based methods such as ChangeFormer [13] and Edge-CVT [1] achieved cutting-edge results by modeling long-range dependencies and improving boundary precision.

Despite innovations, challenges such as edge preservation, misalignment, and intraclass variability continue to have an impact on performance in real-world contexts [14], [15]. The emergence of benchmark datasets such as the large-scale land use change detection dataset (LEVIR-CD+) [16], the Deeply Supervised Image Fusion Network (DSIFN), Wuhan University Change Detection Dataset (WHU-CD), and the Egypt BCD Dataset (EGY-BCD) [1] is critical for progress, as they provide standardized evaluation for complex changes, as shown in Fig. 1. Concurrently, multimodal data fusion and 3-D point cloud analysis present new possibilities [17], [18].

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The authors are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan 430072, China (e-mail: mohamed.elisy8881@gmail.com; xwxiao@whu.edu.cn; muhammed.hatem@bhit.bu.edu.eg; gongjy@whu.edu.cn; drli@whu.edu.cn).

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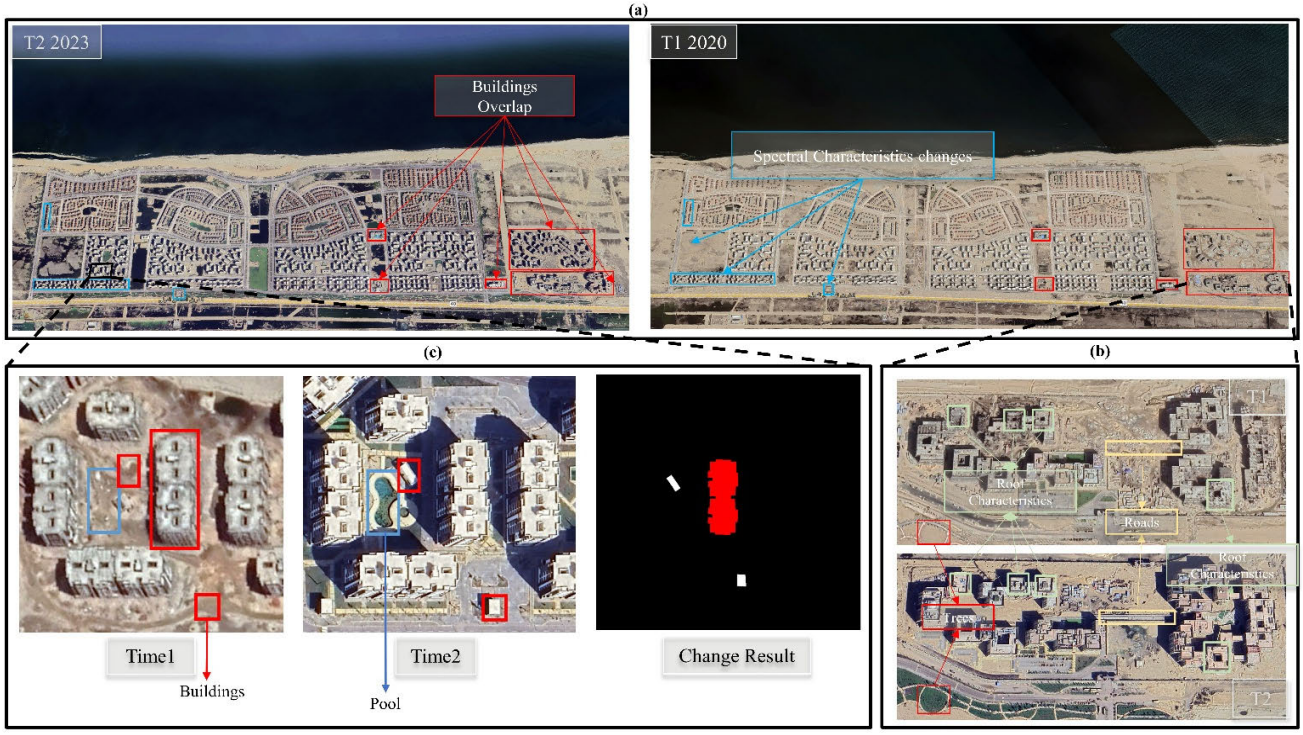


Fig. 1. Example of building change detection in the New Mansoura City. (a) and (b) Bitemporal high-resolution images from 2020 (T1) and 2023 (T2). (c) Detected structural modifications, including new building construction, pool appearance, and changes in roof characteristics. The change result highlights the detected building changes as roof structure changes (red) and new buildings (white) against a black background.

To address these challenges, this study proposes a resolution-standardized change detection network (ReSCD-Net), which directly addresses resolution inconsistency as a first-order problem while developing change detection rather than implicitly recognizing it at the feature level. Unlike previous CNN-, Transformer-, and hybrid-based approaches, which assume fixed input resolutions and rely on scale learning within deep feature representations, ReSCD-Net introduces an adaptive resolution normalizer (ARN) that performs content-aware resolution standardization before segmentation. This framework enables resolution-invariant learning across diverse datasets while keeping precise structural details. The normalized representations are then processed by a resolution-aware segmentation core (RASC), which combines a hierarchical spatial extractor (HSE) and atrous spatial pyramid pooling (ASPP) to capture both fine-grained spatial features and long-term contextual semantics. By explicitly splitting resolution normalization from semantic change modeling, the proposed methodology achieves more robust and exact BCD in a variety of urban contexts and sensor situations. The primary contributions of this study might be summarized as follows.

1) We propose ReSCD-Net, a new resolution-standardized deep learning architecture that directly handles scale inconsistency in bitemporal satellite data by combining an ARN and a single RASC. This architecture allows for consistent, resolution-invariant feature learning across varied datasets, which is not explicitly addressed in existing BCD methods.

2) A content-aware, learnable resolution normalization module is proposed, which standardizes inputs from various spatial resolutions while keeping precise structural and textural characteristics. By completing resolution alignment before segmentation, ARN decreases pseudochanges induced by sensor variability and scale mismatch, boosting robustness and generalization.

3) The provided RASC, which consists of an HSE, an ASPP, and a contextual refinement decoder (CRD), allows for effective multiscale contextual reasoning and precise boundary reconstruction. This approach achieves a balance between global semantic awareness and fine-grained structural delineation, which is critical for accurately detecting building changes in dense, heterogeneous urban landscapes.

The article is organized as follows. Section II presents the related work. Section III describes the proposed ReSCD-Net methodology. Section IV details the datasets and experimental protocol. Section V reports the results and discussion. Finally, Section VI provides the conclusion.

II. Related Work

A. Building Change Detection

BCD is a fundamental task in remote sensing that involves identifying structural changes such as new construction, demolition, or reconstruction between many temporal observations of the same place. It can be used for purposes such as monitoring urban expansion, assessing disaster damage, updating cadastral maps, and managing infrastructure. With the growing availability of high-resolution and very-high-resolution satellite images, BCD has attracted significant interest for its

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capacity to give fine-scale information required for accurate urban analysis. Effective BCD approaches must address issues such as lighting changes, viewpoint variances, seasonal variations, shadows, and pseudochanges, all of which can reduce the accuracy of detected building modifications. This section establishes the problem setting, whereas the following subsections go over the primary categories of methods designed to address these challenges.

B. Traditional Methods

In past decades, conventional image processing methods that predominantly work at the pixel or object level have been applied to BCD. Pixel-based techniques that compare the spectral values of corresponding pixels in bitemporal images to detect changes include image differencing, image ratioing, and change vector analysis [19]. Although computationally efficient, these techniques are very vulnerable to noise, misregistration, and changes in atmospheric conditions and lighting, which can result in a large number of false alarms. In order to address these problems, object-based image analysis (OBIA) was developed. In OBIA, images are first divided into meaningful objects, and then, these objects—rather than individual pixels—are subjected to change detection. Compared to pixel-based methods, OBIA approaches can increase accuracy and decrease "salt and pepper" noise by incorporating spatial, textural, and contextual information. Handcrafted features and classifiers such as random forests (RFs) or support vector machines (SVMs) were frequently used on these objects in early research.

C. CNN-Based Architectures

CNNs have played a critical role in advancing BCD due to their ability to dynamically learn hierarchical, discriminative representations from bitemporal remote sensing images. Classical CNN-based approaches outperformed classic handcrafted-feature methods by allowing end-to-end learning of spatial patterns related to structural change [20]. Early deep learning contributions frequently used semantic segmentation architectures, most notably U-Net, to label each pixel as altered or unchanged, employing encoder–decoder structures capable of preserving both global context and fine spatial information [21]. In addition to encoder–decoder models, many studies have investigated fully convolutional networks for retrieving joint representations from multitemporal inputs. These methods commonly use transfer learning with deep backbones such as the Visual Geometry Group Network (VGG19), while external classifiers such as SVMs were used to construct the final change maps. Siamese CNN architectures, which process T1 and T2 pictures in parallel to build difference-enhancing feature embeddings [22], [21], [23], were particularly prominent. Weight sharing ensures that relevant spatial structures over time are mapped onto a similar feature space, which is critical for modeling minor geometric changes and reducing temporal inconsistencies [19]. Earlier CNN-based models included architectural changes to improve resilience, reduce false alarms, and improve border segmentation. These include residual learning, such as ResNet variations, to speed up gradient propagation [24], [25], attention methods to highlight prominent structural regions, and feature-difference modules that explicitly amplify temporal changes. Other extensions use deep dictionary learning to reduce representation discrepancies between identical objects observed at different times, or they integrate a convolutional long short-term memory network (LSTM) units into U-Net-like models to capture temporal dependencies across multiscale feature hierarchies [26]. More recent approaches have addressed limited annotation settings by utilizing few-shot and metalearning methodologies, allowing for effective model training even under limited supervision.

D. Transformer-Based and Hybrid Architectures

Transformer-based models are attracting a lot of interest for developing change detection because of their capacity to capture long-range dependencies and global contextual relationships, which classic CNNs lack due to their narrow local receptive fields. Vision Transformers (ViTs) use self-attention mechanisms to represent dependencies between distant spatial regions, allowing for a more complete and comprehensive awareness of urban scenes. As a result, multiple studies have shown that transformer-based architectures outperform exclusively CNN-based techniques, especially in densely populated areas where long-range interactions between spatial structures are significant. For instance, the change-aware transformer (CAT) learns generalized representations that highlight altered versus unaffected pixels, which improves temporal feature modeling. Despite these advantages, pure transformer models frequently struggle with fine-grained boundary localization and have a significant computational cost. These restrictions led to the creation of hybrid CNN-Transformer frameworks that take advantage of the beneficial features of both paradigms [27]. Transformer branches capture semantic correlations over large spatial domains, whereas CNN branches enable robust local feature extraction and edge preservation. Previous studies have investigated hybrid systems that integrate ViT modules with CNN encoders to extract high-level semantic cues and generate "change-intensity tokens" for temporal feature interaction. Similarly, the Transformer-CNN Hybrid Network (TCNet) uses adaptive self-attention gating and interactive self-attention in a parallel-branch design to combine multilevel features and bridge semantic gaps across time [28], [29]. Beyond traditional ViT-CNN integration, recent research has also used state space models, such as the Mamba architecture, to provide efficient long-range modeling with much lower computational cost. Hybrid-MambaCD is an ideal instance of combining Mamba-based global modeling with CNN-driven local feature extraction to improve robustness in complicated urban situations. Furthermore, modern hybrid designs use cross-attention techniques to allow direct interaction between bitemporal feature streams, decreasing pseudochanges and enhancing structural consistency in complex circumstances [30]. Recent research has also investigated large-scale foundation models for remote sensing. Thus, SpectralGPT provides a generative pretrained transformer optimized for spectral remote sensing imaging, allowing for flexible handling of different resolutions, spatial extents, and temporal sequences via

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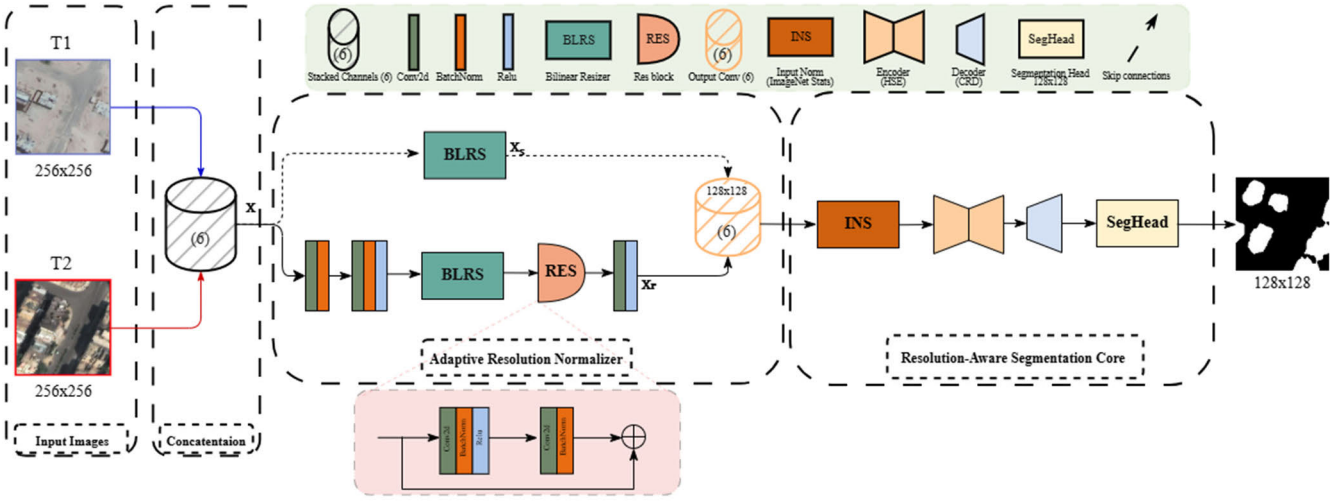


Fig. 2. Framework for the suggested ReSCD-Net model's methodology. The network creates precise building change maps using bitemporal satellite photographs (T1 and T2). The ARN module uses the bilinear resizer (BLRS) and RES to provide content-aware resolution standardization while maintaining spatial consistency and structural fidelity. The normalized features are subsequently processed by the RASC, which consists of INS, the HSE encoder, and the CRD. Finally, the SegHead generates high-resolution binary change masks using BSL to ensure accurate delineation of building boundaries and prevent false detections.

large-scale self-supervised pretraining [31]. In parallel, foundation segmentation models have been developed for urban contexts. UrbanSAM builds on the segment anything model by including a learnable scale-adaptive prompter inspired by multiresolution analysis, resulting in better segmentation of scale-varying urban items [32]. The Cross-City Matters benchmark uses large-scale multimodal datasets and a domain adaptation framework (HighDAN) to address performance degradation in diverse urban environments [33]. While such foundation models have great generalizability across different downstream tasks, their high computing cost and data needs make them unsuitable for task-specific, high-resolution BCD scenarios.

E. Limitations and Open Challenges

Even with great advancements, there are still a number of restrictions and unresolved issues that prevent BCD from being widely used in practice [34], [35]. Distinguishing real building changes from pseudochanges brought on by a variety of circumstances, including seasonal variations, variations in lighting and shadows, and misregistration between multitemporal images, is one of the main challenges [23], [36], [37]. According to [2], these elements may result in geometric distortions and radiometric irregularities, which raise the frequency of false positives (FPs). The lack of large-scale, high-quality annotated datasets is another crucial problem that significantly restricts the practical applicability of deep learning models, which usually call for enormous quantities of training samples [38], [39]. Furthermore, it is difficult to accomplish precise edge detection and accurate building outline extraction in urban environments due to the complexity and wide variety of building shapes and sizes [23], [40], [41]. In order to improve detection accuracy, research is still being done on how to best utilize temporal–spatial correlations and achieve good discrimination in the vicinity of building edges [40], [42]. CNN-based methods excel at local feature extraction but struggle with global context and resolution variability, while transformer-based approaches improve long-range modeling at the expense of increased computing complexity and reduced boundary precision. Hybrid architectures help to address some of these challenges, but they still rely on fixed-resolution inputs and are susceptible to scale differences between datasets. As a result, conventional techniques prioritize feature-level adaptation above tackling resolution inconsistency as a first-order problem. This gap inspired the proposed resolution-aware approach, which directly normalizes spatial scale before segmentation and enhances multiscale contextual reasoning with increased structural consistency.

III. Methodology

A. Model Overview

We present ReSCD-Net, a trainable BCD pipeline that combines RASC with ARN in an end-to-end framework. The ARN improves feature representation and enhances spatial resolution by concatenating bitemporal satellite pictures along the channel dimension, resulting in a six-channel input ($t_1 \oplus t_2$). The RASC, which is based on a DeepLabV3+ structure and an EfficientNet-B4-based HSE, normalizes the outputs and generates pixelwise binary change masks. Fig. 2 shows the pipeline as a whole. This architecture has three primary advantages: 1) it enables resolution-agnostic training across heterogeneous datasets; 2) it maintains a balance between global context modeling and fine-grained boundary preservation; and 3) it guarantees robustness in detecting subtle structural modifications, such as rooftop expansions or small-scale demolitions.

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B. Adaptive Resolution Normalizer

Inspired by [43], the ARN is a content-aware resizing module that unifies multiple spatial resolutions while keeping semantically essential data. Unlike fixed interpolation techniques such as bilinear or bicubic resampling, ARN uses a trainable CNN-based residual architecture that includes batch normalization, LeakyReLU activations, and residual convolutional blocks. This technique enables adaptive, data-driven scaling while preserving fine-grained structural features required for BCD. The module receives six-channel inputs of size 256×256 pixels and produces uniformly resized outputs of 128×128 pixels. This transformation, shown in the ARN block of Fig. 2, is not a naive geometric operation but a learned process that preserves semantic boundaries, suppresses background noise, and corrects minor spatial misalignments. The output is fully compatible with the segmentation model via a skip-add mechanism enabled by matched input/output channel dimensions. Structurally, it is composed of strided convolutions for downsampling after several convolutional layers with residual connections. LeakyReLU activations and batch normalization are used to maintain nonlinear feature variations and stabilize training. This module handles the concatenated six-channel picture as input ($I \in R^{H\times W\times C}$), where $C = 6$, and outputs a resized representation $I' \in R^{H'\times W'\times C'}$. The convolutional residual blocks (RESs) allow the model to emphasize edges, corners, and texture features critical for detecting structural changes. The corresponding calculation formulas for this process are provided as follows:

$$F_l = \sigma\left(\mathrm{BN}\left(W_l * F_{l-1} + b_l\right)\right) + F_{l-1} \tag{1}$$

where F_{l-1} and F_l denote the input and output feature maps at the lth RES, W_l and b_l are the convolutional weights and biases, BN(·) is batch normalization, and $\sigma(\cdot)$ represents the LeakyReLU activation. The final resizing is achieved using a stride convolution, defined as

$$I'_{x,y} = \sum_{i=1}^{k}\sum_{j=1}^{k} w_{i,j} \cdot I_{s\cdot x+i,s\cdot y+j} \tag{2}$$

where k is the kernel size, s is the stride, and $w_{i,j}$ are learnable filter parameters. The resizer is guaranteed to learn to maintain semantically significant features such as building borders and rooftop boundaries in addition to standardizing the resolution across datasets, thanks to this architecture.

The ARN uses this learned resizing technique to enable resolution-agnostic training, cross-dataset standardization, and semantic consistency among multitemporal images. Its adaptive input normalization (INS) allows the downstream RASC to function on uniformly scaled, feature-rich representations, significantly enhancing robustness and generalization over a wide range of satellite sources and imaging conditions.

C. Resolution-Aware Segmentation Core

The RASC, shown in Fig. 2, is the key component of the proposed ReSCD-Net framework. The RASC is based on a DeepLabV3+ segmentation backbone, which is generated using the SMP library and includes a timm-EfficientNet-B4 encoder that has been pretrained on ImageNet to initialize the HSE. The model is optimized for six-channel inputs, representing stacked bitemporal imaging ($T_1 \oplus T_2$). The first convolutional layer is patched to handle the increased input depth. Normalization is performed using duplicated ImageNet statistics to ensure that feature distributions are constant across temporal channels. RASC aims to effectively capture fine-grained structural information and wide contextual dependencies, both of which are required for accurate BCD. DeepLabV3+ demonstrated superior performance in multiscale semantic segmentation tests [44], making it ideal for this application, which requires edge precision and spatial coherence. There are three primary parts to the network.

1) HSE is an encoder based on EfficientNet-B4 that uses squeeze-and-excitation (SE) blocks and depthwise separable convolutions to extract hierarchical spatial representations effectively. This program detects localized structural elements like building edges and rooftops while being computationally efficient.

2) ASPP uses parallel dilated convolutions with different dilation rates to capture multiscale context.

3) CRD is a lightweight decoder that combines high-level semantic information from the ASPP with low-level encoder outputs to refine object boundaries and improve the delineation of subtle structural variations. As shown in Fig. 2, the model produces a single-channel binary mask, representing changed versus unchanged regions. By combining multiscale context abstraction, edge-preserving upsampling, and the semantically enhanced 128×128 features provided by the ARN, the RASC model achieves high accuracy and robustness across diverse datasets and imaging conditions.

D. Encoder: HSE

The HSE, shown in Fig. 3, serves as the encoder component of the RASC module, gradually extracting hierarchical feature representations from resolution-normalized bitemporal inputs. Built on the EfficientNet-B4 foundation, HSE uses SE blocks and depthwise separable convolutions to strike a compromise between representational power and processing economy. The SE mechanism adjusts channelwise feature responses by highlighting structurally important activations, such as building edges, rooftops, and façade textures, increasing sensitivity to small spatial changes. Meanwhile, depthwise separable convolutions greatly reduce the amount of parameters and floating-point operations (FLOPs), allowing for fast large-scale inference while preserving spatial integrity.

The HSE downsamples the input images gradually over a series of convolutional stages while enhancing the semantic depth of derived features. The resulting multilevel feature maps $(F_1, F_2, \ldots, F_n)$ record both fine-grained geometric information and high-level contextual semantics, important for spotting structural changes over time. These encoded features are then passed into the ASPP and CRD for additional aggregation and boundary refinement. The HSE, which combines parameter-efficient convolutional processes with adaptive channel attention, provides a compact yet effective feature extraction technique that ensures ReSCD-Net's correctness and scalability across multiresolution satellite datasets.

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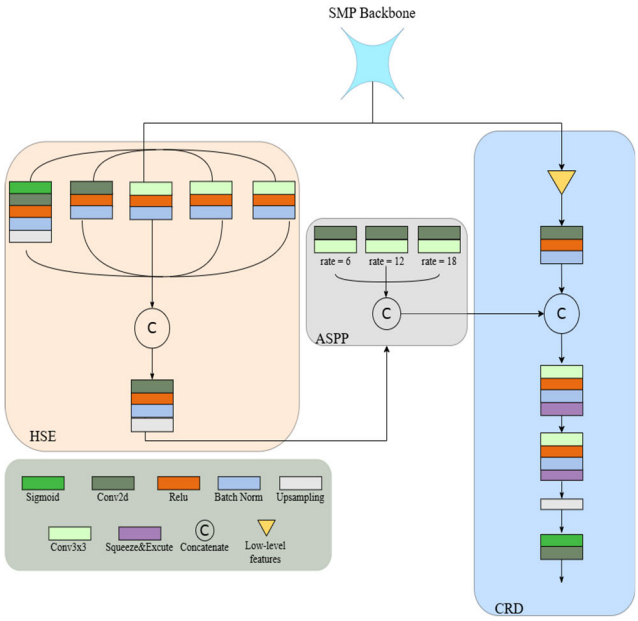


Fig. 3. Detailed structure of the RASC, including the HSE, ASPP, and the CRD. The encoder extracts hierarchical features, the ASPP module enriches multiscale context, and the decoder refines and upsamples feature maps to produce precise building change masks. C denotes concatenation, and the triangle denotes upsampling.

E. Atrous Spatial Pyramid Pooling

In the RASC depicted in Fig. 2, the ASPP module is used following the HSE to improve multiscale contextual reasoning before the features are given to the CRD. By using parallel atrous convolutions with multiple dilation rates, the ASPP enriches the semantic representation, allowing the network to capture both broad contextual patterns, including block-scale urban growth, and fine-grained structural details, including small rooftop modifications and partial demolitions. The corresponding calculation formulas for this process are provided as follows:

$$F_{\text{ASPP}} = \text{Concat}\left(\text{Conv}_{1\times1}(F), \text{Conv}_{3\times3,r_1}(F) \times \text{Conv}_{3\times3,r_2}(F), \text{Conv}_{3\times3,r_3}(F), \text{Pool}\right) \quad (3)$$

where F denotes the encoder output, $r_1, r_2,$ and r_3 represent dilation rates, and Pool$(\cdot)$ refers to global average pooling. The outputs are concatenated to form a feature map enriched with multiscale contextual information.

F. Decoder: CRD

The CRD, shown in Fig. 3, is intended to refine the coarse change representations produced by the HSE and ASPP modules of the RASC. It does multilevel feature fusion by combining low-level spatially rich encoder characteristics. $\boldsymbol{F_{low}}$, which retain fine spatial details, with the high-level semantic features $\boldsymbol{F_{ASPP}}$. This hierarchical fusion improves both boundary precision and situational awareness, which are critical for accurately detecting building changes in complicated urban environments. The CRD uses a skip-connection fusion method, followed by convolutional refinement layers, to recreate fine-grained information that may have been lost during encoding. This architecture allows the network to reconstruct building outlines and structural boundaries, ensuring accurate differentiation between authentic and virtual alterations. The formulas for the decoder output are given as follows:

$$P = \sigma\left(W_d * \text{Concat}\left(F_{\text{ASPP}}, F_{\text{low}}\right) + b_d\right) \quad (4)$$

where $P \in [0,1]^{H'\times W'}$ is the predicted probability map, W_d and b_d are learnable decoder parameters, and $\sigma(\cdot)$ is the sigmoid activation for binary classification.

By combining spatial detail preservation from lower-level features with semantic abstraction from higher layers, the CRD enhances boundary localization and object coherence, making it especially efficient at detecting fine-scale building modifications and rooftop expansions across heterogeneous scenes.

G. Loss Function: BSLs

The balanced structural loss (BSL) is used as an optimization criterion to direct the network toward precise and balanced BCD. It is based on the pixelwise binary cross-entropy loss, which is frequently used in semantic segmentation and change detection applications [12], [13]. BSL ensures that every pixel in the projected probability map is correctly identified as "change" or "no change," which reinforces fine structural localization and reduces false detections. Given the major resolution of satellite photography, even minor spatial misclassifications can result in large structural inconsistencies in crowded metropolitan regions. To overcome this, BSL uses pixel-level supervision, which penalizes inaccurate projections more severely in areas with complex geometry, such as building margins or thin rooftop structures. The network produces a probability map $P \in [0,1]^{H\times W}$, where each element p_i represents the likelihood of change at pixel i and the corresponding ground-truth label is denoted as $y_i \in \{0,1\}$. The BSL for binary classification is expressed as follows:

$$L_{\text{BSL}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log\left(1-p_i\right)\right] \quad (5)$$

where $N = H \times W$ is the total number of pixels in the input.

This approach assures equal gradient propagation for both changed and unchanged classes, reducing the impact of class imbalance, a common problem in change detection where unaltered pixels predominate. By enforcing structural consistency and balanced optimization, BSL improves the network's ability to preserve fine boundaries while maintaining global change coherence, effectively complementing the CRD's spatial learning.

IV. Datasets and Experimental Analysis

A. Datasets

The change detection pipeline is evaluated using a diverse set of bitemporal and multitemporal datasets, each capturing distinct types of semantic and structural changes. As the datasets differ in resolution and format, all images are uniformly cropped into 256×256 tokens to enable consistent input for downstream processing. This setup supports robust evaluation of generalization, temporal alignment, and noise

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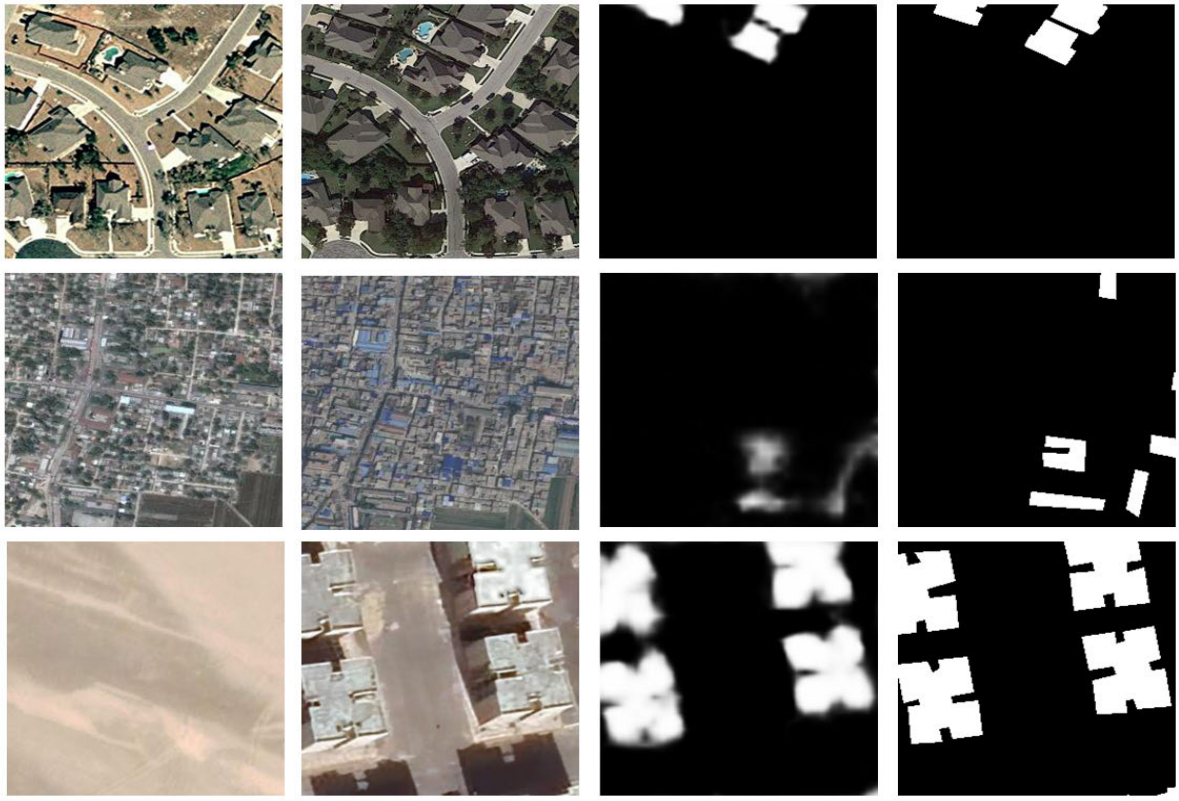


Fig. 4. Visual examples of BCD across datasets. Each row shows a bitemporal image pair (left two columns), the predicted change map by ReSCD-Net (third column), and the corresponding ground-truth mask (right column). From (top to bottom): examples from LEVIR-CD+, DSIFN-CD, and EGY-BCD datasets, illustrating the model's ability to detect diverse structural changes in suburban, dense urban, and desert expansion scenes.

TABLE I

EXTENSIVE DETAILS ABOUT THE EXPERIMENTAL DATASET USED IN THIS STUDY, INCLUDING ITS CHARACTERISTICS, PARTITIONING STRATEGIES, AND KEY ATTRIBUTES

Dataset	Image Size (pixels)	Resolution (m)	Image Pairs	Bands	Train	Val	Test	Data Collection
EGY-BCD	256 × 256	0.25	6,091	RGB	4,263	1,218	610	Google Earth
LEVIR-CD+	1024 ×1024	0.5	985		637	128	220	Google Earth
DSIFN-CD	512 × 512	2.0	4,000		2,000	1,000	1,000	Google Earth

resilience across varied real-world scenarios. To evaluate the effectiveness of the proposed framework, experiments were conducted on three publicly available benchmark datasets that represent diverse geographic and structural contexts: EGY-BCD [1], LEVIR-CD+ [16], and DSIFN-CD [45], as detailed in Fig. 4. Fig. 4 shows an example of the input images before and after the change (T_1 and T_2), as well as the change masks and boundary labels. Table I provides specific information about the datasets, which are further described in the following.

1) *EGY-BCD:* The EGY-BCD dataset comprises 6091 bitemporal image pairs at 256×256 pixels and 0.25-m spatial resolution, sourced from Google Earth from two different time periods (2015–2022). Since the images are already in the required resolution, no patching or cropping is applied, and the dataset remains at 6091 full-size samples. It is dedicated to detecting building-level changes in Egyptian urban environments, using only RGB bands, and is divided into 4263 training, 1218 validation, and 610 test samples. The dataset emphasizes precision in dense urban areas with variable architectural styles.

2) *LEVIR-CD+:* The LEVIR-CD+ dataset provides 985 high-resolution bitemporal image pairs, each sized 1024 × 1024 pixels at 0.5-m resolution, sourced from Google Earth. Designed for semantic BCD, it captures diverse urban expansion scenarios. As part of preprocessing, each image pair is divided into nonoverlapping 256 × 256 patches, resulting in a total of 15 760 cropped image tokens. The dataset contains only RGB bands, with 637 samples used for training and 348 for testing, making it ideal for benchmarking models on large, high-resolution scenes with consistent semantics.

3) *DSIFN-CD:* The DSIFN-CD dataset includes 3988 bitemporal image pairs of 512×512 pixel resolution, covering dense urban, industrial, and rural landscapes in China. It

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focuses on binary BCD with rich geographical diversity. All images are partitioned into four 256×256 patches per pair, yielding a total of 15 952 cropped image tokens. Only RGB bands are provided. The dataset's heterogeneity makes it well-suited for testing model generalization across diverse city morphologies and construction patterns.

B. Comparison Methods

We evaluate the effectiveness of the proposed model by comparing its results with those of nine representative and cutting-edge BCD models. These baselines offer a wide range for assessment since they cover both recent transformer-enhanced architectures and early convolutional techniques. These techniques cover a wide range of architectures, from state-of-the-art transformer and edge-aware methods to traditional convolutional Siamese designs. A complete and equitable assessment of the suggested model's capacity to strike a balance between boundary accuracy, contextual awareness, and computational efficiency is made possible by benchmarking against them.

1) *DSAM-Net [46]:* A dual-stream attention-based network that aims to detect changes by capturing complementary spatial and semantic features. It increases the bitemporal representations' capacity for discrimination by utilizing attention mechanisms.

2) *FC-Siam-Conc [12]:* A fully convolutional Siamese model that simultaneously processes images taken before and after an event, concatenating their features to classify them. For tasks involving bitemporal change detection, it offers a solid baseline.

3) *FC-Siam-Diff [12]:* It enables the direct modeling of temporal differences between the two images by using feature differencing rather than concatenation.

4) *BIT-Net [47]:* It introduces self-attention mechanisms to enhance feature alignment for the detection of change at the urban scale by better capturing long-range dependencies over time.

5) *CTFI-Net [48]:* A cross-temporal feature interaction network that improves the detection of minor structural changes by focusing on fine-grained semantic correspondence between multitemporal images.

6) *ChangeFormer [13]:* A Siamese structure based on transformers featuring self-attention modules for modeling global context. By striking a balance between local detail and long-range dependencies, it exhibits competitive results.

7) *ICIF-Net [49]:* An interactive cross-temporal information fusion network that enhances temporal correlation modeling through multilevel feature aggregation and semantic refinement.

8) *EGCT-Net [50]:* A CNN–Vision Transformer hybrid framework that employs edge supervision and transformer-based feature modeling. Maintaining boundaries and reducing spectral interference, it sets a high bar for high-resolution urban imagery.

9) *Edge-CVT [1]:* A CNN–Vision Transformer hybrid framework that utilizes transformer-based feature modeling and edge-guided. It establishes a high standard for high-resolution urban imagery by maintaining boundaries and minimizing spectral interference.

10) *DTCDSCN [51]:* A deep Siamese convolutional network with dual-task restrictions that learns both change detection and semantic representation. By applying task-level constraints, it enhances discrimination of altered regions while retaining structural consistency.

11) *SNUNet [52]:* A densely linked Siamese network optimized for very high-resolution (VHR) imaging. It uses layered skip connections to improve multi-scale feature propagation and record finegrained changes in complicated urban settings.

C. Implementation Details

All analyses were carried out on a dedicated deep learning workstation outfitted with an NVIDIA RTX 3090 GPU (24-GB VRAM), an Intel Xeon processor with 16 CPU cores, 64-GB RAM, and a 500-GB NVMe SSD to provide fast data access and efficient model training. The machine operated Ubuntu 20.04 LTS, and all models were built in PyTorch 1.12 with CUDA 11.6 support. Mixed-precision training (FP16) was used to reduce memory consumption and speed up calculations. We utilized the Adam optimizer with an initial learning rate of 1×10^{-4}, a weight decay of 1×10^{-5}, and an 8-batch size. Data augmentation included random orientation flips, rotations, and modest brightness-contrast jittering to improve robustness under a variety of imaging situations. During training, all bitemporal image pairs were cropped into 256×256 patches. The segmentation head (SegHead) generated 128×128 change masks based on the CRD decoder's output stride. This arrangement provided consistent GPU utilization, reproducible experimental performance, and assessment across the EGY-BCD, LEVIR-CD+, and DSIFN-CD datasets.

D. Evaluation Metrics

The evaluation's objective is to compare performance to the most advanced techniques while examining the model's advantages and disadvantages in various situations. Well-known binary change detection metrics, such as precision, recall, $F1$-score, Intersection over Union (IoU), mean IoU (mIoU), and overall accuracy (OA), are utilized for evaluating performance. These metrics capture pixel-level accuracy, spatial consistency, and the tradeoff between FPs and false negatives (FNs). These metrics are defined as follows:

$$\begin{aligned} \text{Precision} &= \text{TP}/(\text{TP}+\text{FP}) \\ \text{Recall} &= \text{TP}/(\text{TP}+\text{FN}) \\ F1 &= 2 * \text{Precision} * \text{Recall}/(\text{Precision}+\text{Recall}) \\ \text{IoU} &= \text{TP}/(\text{TP}+\text{FP}+\text{FN}) \\ \text{mIoU} &= \left(\text{IoU_change} + \text{IoU_no-change}\right)/2 \\ \text{OA} &= (\text{TP}+\text{TN})/(\text{TP}+\text{TN}+\text{FP}+\text{FN}) \end{aligned} \tag{6}$$

where FP represents false positives, FN indicates false negatives, TN denotes true negatives, and TP refers to true positives.

V. Results and Discussion

This section uses the standard metrics (precision, recall, $F1$, mIoU, and OA) to report quantitative performance on LEVIR-CD+, DSIFN-CD, and EGY-BCD. The suggested

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TABLE II

RESULTS ON THE EGY-BCD DATASET. BASELINE RESULTS ARE MAINLY ADOPTED FROM THE ORIGINAL EDGE-CVT [1] PUBLICATION AND THEIR REFERENCED SOURCES, WHILE THE RESULTS OF RESCD-NET ARE OBTAINED IN THIS WORK. THE BEST RESULTS ARE HIGHLIGHTED IN RED, WHILE THE SECOND-BEST ARE IN BOLD

Method	Precision	Recall	F1	mIoU	OA
DSAM-Net (2021) [46]	88.58%	71.50%	79.13%	69.50%	95.79%
FC-Siam-Conc (2018) [12]	82.77%	85.56%	84.14%	75.05%	95.88%
FC-Siam-Diff (2018) [12]	82.95%	86.48%	84.68%	75.72%	95.98%
BIT-Net (2021) [47]	**91.30%**	82.65%	86.76%	78.50%	96.96%
CTFI-Net (2023) [48]	89.80%	84.69%	87.17%	80.37%	97.17%
ChangeFormer (2022) [13]	84.71%	**91.88%**	88.15%	80.78%	97.15%
ICIF-Net (2022) [49]	89.52%	87.42%	88.46%	81.47%	97.29%
EGCT-Net (2022) [50]	89.77%	88.70%	89.23%	82.57%	97.35%
Edge-CVT (2025) [1]	**90.17%**	90.07%	**90.12%**	**83.96%**	**98.13%**
ReSCD-Net(Ours)	88.79%	**93.48%**	**91.07%**	**91.12%**	**98.72%**

TABLE III

RESULTS ON THE LEVIR-CD+ TEST SET. BASELINE RESULTS ARE MAINLY ADOPTED FROM THE ORIGINAL EDGE-CVT [1] PUBLICATION AND THEIR REFERENCED WORKS, WHILE THE RESULTS OF RESCD-NET ARE OBTAINED IN THIS STUDY. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST IN BOLD

Method	Precision	Recall	F1	mIoU	OA
DSAM-Net (2021) [46]	87.74%	88.93%	88.33%	79.89%	92.99%
FC-Siam-Conc (2018) [12]	89.85%	90.57%	90.21%	82.75%	94.16%
FC-Siam-Diff (2018) [12]	90.70%	90.16%	90.43%	82.89%	94.29%
ChangeFormer (2022) [13]	92.38%	91.50%	91.93%	85.54%	95.28%
CTFI-Net (2023) [48]	92.72%	91.89%	92.29%	86.09%	95.49%
EGCT-Net (2022) [50]	92.62%	**92.84%**	92.73%	86.47%	95.59%
BIT-Net (2021) [47]	95.39%	91.82%	**93.58%**	87.36%	95.69%
ICIF-Net (2022) [49]	95.61%	92.28%	93.41%	87.94%	95.98%
Edge-CVT (2025) [1]	**96.85%**	91.81%	**94.26%**	**89.79%**	**96.73%**
ReSCD-Net(Ours)	**98.54%**	**93.36%**	93.06%	**90.83%**	**99.02%**

ARN and RASC method ("Our work") is listed alongside recent baselines from the literature in each dataset (see Tables II–V). The objective is to present a straightforward, comparative analysis of detection reliability and segmentation quality across a range of scenes and resolutions.

A. Experiments on EGY-BCD Dataset

The results of the evaluation of the suggested framework on the EGY-BCD test set are compiled in Table II. The performance parameters of baseline approaches are taken from the original Edge-CVT [1] publication, which collects findings from multiple studies, to ensure consistency and fair comparison. The proposed ReSCD-Net results are computed separately in this work. The model exhibits good overall balance on scenes centered around Egyptian buildings, with an $F1$-score of 91.09%, a recall of 93.48%, and an OA of 98.72%. These outcomes outperform a number of cutting-edge baselines, most notably Edge-CVT [1] in both $F1$ (90.12%) and recall (90.07%). The ARN ensures cross-dataset resolution consistency by harmonizing high-detail (0.25 m) imagery, while RASC, with its multiscale contextual reasoning, enhances the delineation of building boundaries and structural integrity, as shown in Fig. 5. The visual comparison clearly shows that the proposed method yields fewer FNs and FPs than other approaches such as ChangeFormer, BIT-Net, and DSAM-Net. In contrast to previous Siamese-based architectures such as FC-Siam-Conc and FC-Siam-Diff [12], the suggested approach produces gains of roughly +6.9% and +6.4% in $F1$, respectively, highlighting the advantages

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TABLE IV

RESULTS ON THE DSIFN-CD TEST SET. THE BASELINE RESULTS ARE PRIMARILY BASED ON THE ORIGINAL CHANGEFORMER PUBLICATION [13] AND ITS REFERENCES, WHEREAS THE RESULTS OF THE PROPOSED TECHNIQUE ARE OBTAINED IN THIS STUDY. THE BEST OUTCOMES ARE DISPLAYED IN RED, WHILE THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Precision	Recall	F1	IoU	OA
FC-Siam-Diff (2018) [12]	59.67%	65.71%	62.54%	45.50%	86.63%
FC-Siam-Conc (2018) [12]	66.45%	54.21%	59.71%	42.56%	87.57%
FC-EF [12]	72.61%	52.73%	61.09%	43.98%	88.59%
DTCDSCN [51]	53.87%	77.99%	63.72%	46.76%	84.91%
SNUNet [52]	60.60%	72.89%	66.18%	49.45%	87.34%
BIT-Net (2021) [47]	68.36%	70.18%	69.26%	52.97%	89.41%
ChangeFormer (2022) [13]	**88.48%**	**84.94%**	**86.67%**	**76.48%**	**95.56%**
ReSCD-Net(Ours)	**92.89%**	**93.76%**	**93.32%**	**87.48%**	**97.72%**

TABLE V

RUNNING EFFICIENCY AND PARAMETERS WITH $F1$-SCORE ON THE EGY-BCD DATASET. THE COUNTS ARE ADOPTED FROM THE ORIGINAL EDGE-CVT ARTICLE [1], AND THE QUOTED $F1$-SCORES INCLUDE RESULTS FROM THIS STUDY

Model	FLOPs (G)	#Parameters (M)	F1-score (%)
DSAM-Net (2021) [46]	603.10	16.95	79.13
FC-Siam-Conc (2018) [12]	42.65	1.55	84.14
FC-Siam-Diff (2018) [12]	37.81	1.35	84.68
BIT-Net (2021) [47]	70.02	3.04	86.76
CTFI-Net (2023) [48]	100.51	4.19	87.17
ChangeFormer (2022) [13]	1640.00	46.64	88.15
ICIF-Net (2022) [49]	203.28	23.84	88.46
EGCT-Net (2022) [50]	307.78	106.13	89.23
Edge-CVT (2025) [1]	297.12	95.77	90.12
ReSCD-Net(Ours)	**120.60**	**35.20**	**91.07**

of combining robust preprocessing with multiscale contextual reasoning.

In comparison to BIT-Net [47] and CTFI-Net [48], which attain $F1$-scores of 86.76% and 87.17%, respectively, the suggested framework exhibits gains of more than +4% points, indicating a better capacity to detect minute building changes. Even though precision (88.79%) is marginally less than edge-focused techniques such as Edge-CVT (90.17%), this tradeoff makes sense, given that the current implementation lacks explicit edge supervision. However, a more balanced treatment of change and no-change classes is suggested by the higher mIoU (91.12%) in comparison to Edge-CVT (83.96%), which improves reliability in crowded urban areas. In addition, while models such as ICIF-Net [49] and ChangeFormer [13] demonstrate competitive recall, their overall $F1$ and OA scores are still lower than the suggested method, confirming its efficacy in Egyptian urban environments. According to these results, the suggested framework works especially well in high-resolution, diverse urban settings where different architectural styles and congested building configurations frequently pose problems for current models. The findings also point to potential avenues for future development, such as incorporating region-level calibration methods, edge-aware loss functions, and confidence-guided thresholding to increase accuracy while preserving high recall.

B. Experiments on the LEVIR-CD+ Dataset

The proposed ReSCD-Net achieves an $F1$-score of 93.06%, an mIoU of 90.83%, and an extraordinarily high OA of 99.02% on the LEVIR-CD+ benchmark, as shown in Table III. The baseline results are mainly adopted from the original Edge-CVT [1] publication and its referenced sources to ensure fair and consistent comparison, whereas the results of the proposed method are independently obtained in this study. As illustrated in Fig. 6, ReSCD-Net produces cleaner change maps with fewer FNs and FPs than conventional CNN- and transformer-based models, effectively preserving building boundaries and detecting subtle rooftop modifications. ReSCD-Net outperforms traditional Siamese CNN baselines such as FC-Siam-Conc (90.21% $F1$) and FC-Siam-Diff (90.43% $F1$) [12], demonstrating the effectiveness of resolution-agnostic preprocessing via the ARN and enhanced contextual modeling with ASPP. It also outperforms DSAM-Net (88.33% $F1$) [46], indicating the architecture's robustness in handling scale and scene diversity. Although transformer-based and hybrid designs such as ChangeFormer (91.93%

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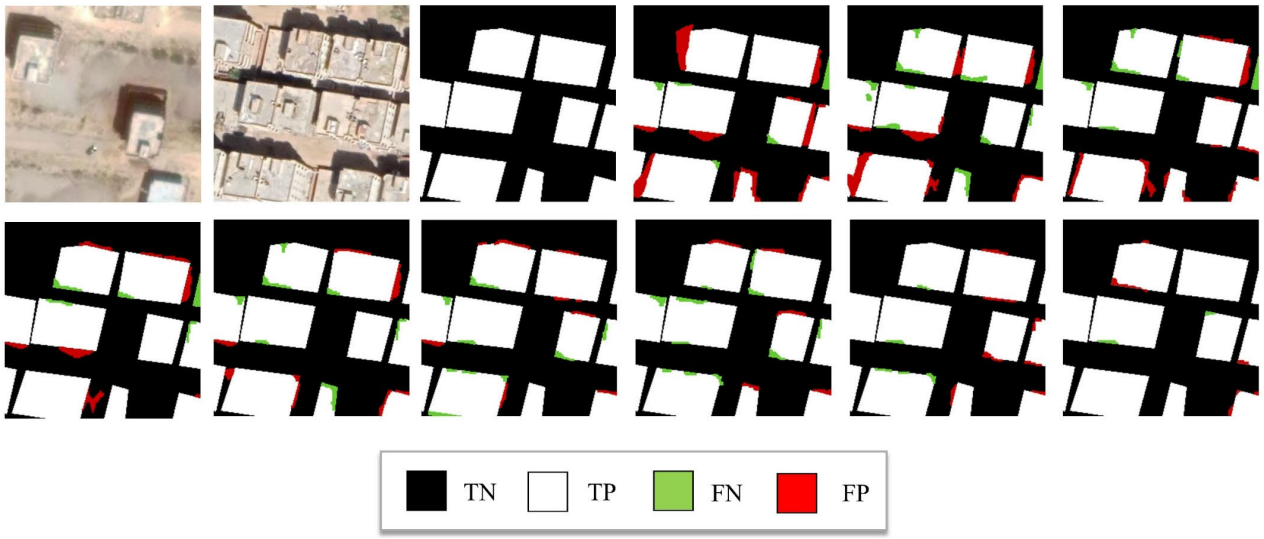


Fig. 5. Visualization comparison of change detection results on the EGY-BCD dataset. From (left to right): (a) T1 image and (b) T2 image, ground truth label, and the predictions of BIT-Net, ChangeFormer, DSAM-Net, EGCT-Net, FC-Siam-Conc, FC-Siam-Diff, ICIF-Net, Edge-CVT, and ReSCD-Net (Ours). TPs are shown in white, TNs in black, FNs in green, and FPs in red.

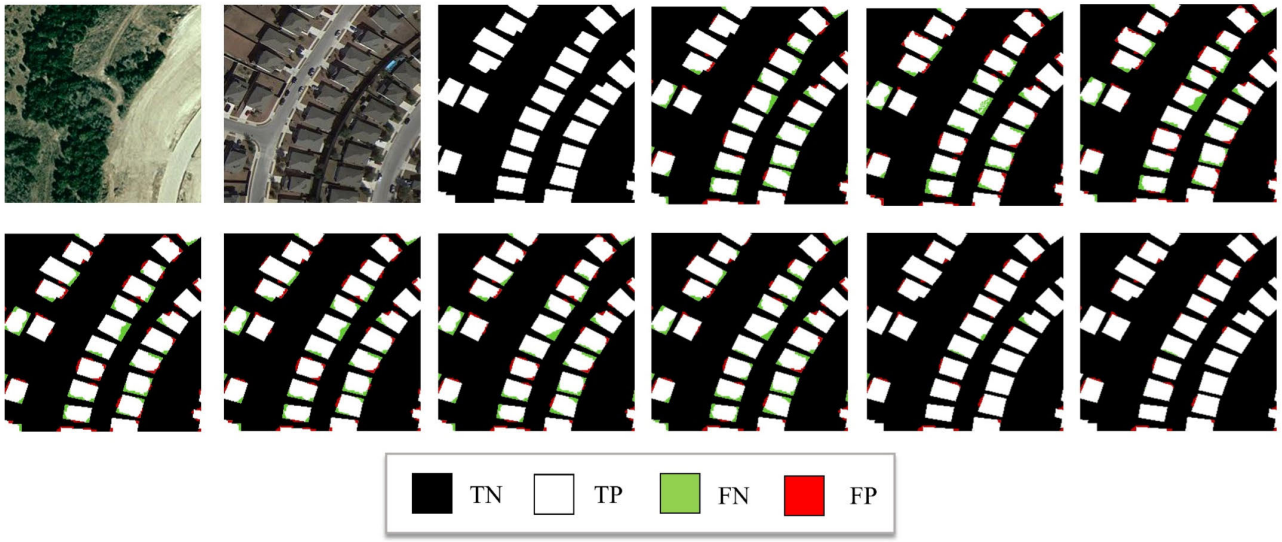


Fig. 6. Visualization comparison of change detection results on the LEVIR-CD+ dataset. From left to right: (a) T1 image and (b) T2 image, ground truth label, and the predictions of BIT-Net, ChangeFormer, DSAM-Net, EGCT-Net, FC-Siam-Conc, FC-Siam-Diff, ICIF-Net, Edge-CVT, and ReSCD-Net (Ours). TPs are shown in white, TNs in black, FNs in green, and FPs in red.

$F1$), CTFI-Net (92.29% $F1$), and EGCT-Net (92.73% $F1$) perform well, ReSCD-Net outperforms them by getting a higher $F1$-score (93.06%) and mIoU (90.83%). Compared to more edge-focused baselines, the model has excellent recall (93.36%) and balanced segmentation accuracy, but precision (98.54%) is slightly higher than approaches such as ICIF-Net (95.61% precision and 93.91% $F1$) and Edge-CVT (96.85% precision and 94.26% $F1$) [1]. This improvement demonstrates that, while explicit boundary refinement in edge-aware designs sharpens contours, ReSCD-Net's scale-normalization technique produces competitive results in the absence of dedicated edge modules. The remarkable OA (99.02%) is contributed to the LEVIR-CD+ class imbalance, which favors unchanging pixels. As a result, $F1$ and mIoU remain the most accurate markers of balanced detection performance. Overall, the results show that ReSCD-Net is not only competitive with previous Transformer-based and edge-enhanced models, but it also has higher recall and robustness in dense urban situations, making it suitable for practical BCD applications.

C. Experiments on the DSIFN-CD Dataset

As shown in Table IV, our model establishes a clear state-of-the-art among the listed methods. The baseline results on the DSIFN-CD dataset are primarily based on the original ChangeFormer paper [13], which offers comparison results across several approaches, whereas the results of the suggested strategy are computed separately in this work. Our method delivers $F1$ = 93.32% and IoU = 87.48%, surpassing ChangeFormer ($F1$ = 86.67% and IoU = 76.48%) by +6.65 and +10.84 points, respectively. DSIFN-CD aggre-

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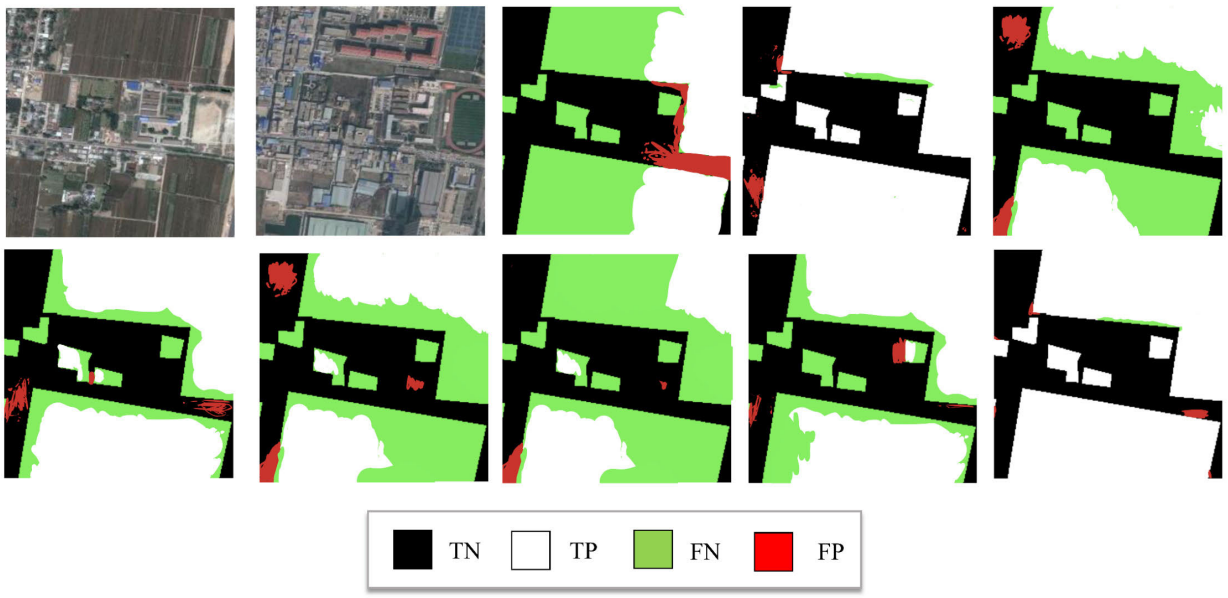


Fig. 7. Visualization of BCD on the DSIFN-CD dataset. From (top left to bottom right): (a) preevent and (b) postevent, BIT-Net, ChangeFormer, DSAM-Net, EGCT-Net, FC-Siam-Conc, FC-Siam-Diff, ICIF-Net, and OURS (ReSCD-Net). TPs are highlighted in white, TNs in black, and misclassified regions represent FPs and FNs. The suggested ReSCD-Net achieves higher border precision while reducing false detections.

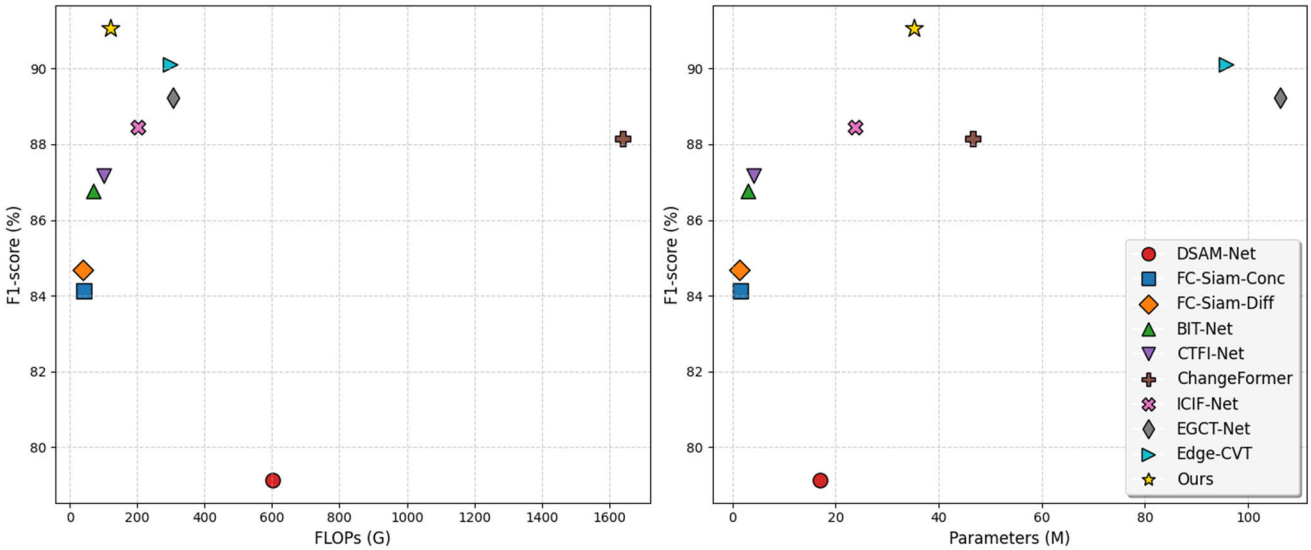


Fig. 8. Comparison of model complexity using the EGY-BCD dataset for different approaches in terms of parameters, computing cost (FLOPS), and $F1$-score values.

gates diverse urban/industrial contexts at 512×512, making cross-resolution consistency and scale robustness critical. The ARN standardizes inputs (256 to 128×128 features), while the RASC captures multiscale cues, jointly improving recall (93.76%) without sacrificing precision (92.89%). The simple morphology-based postprocessing is sufficient here because DSIFN's changes are typically mesoscale (blocks and roofs), where context aggregation dominates edge microdetail. As shown in Fig. 7, the visual comparison clearly demonstrates that ReSCD-Net efficiently delineates structural boundaries while decreasing false activations. Several existing models, such as FC-Siam-Diff, BIT-Net, and DSAM-Net, either ignore background clutter or disregard subtle building transformations. ReSCD-Net's predictions preserve sharp edge consistency while eliminating false alarms around small-scale urban features such as industrial blocks and roof alterations. TPs, outlined in white, are noticeably more continuous and better aligned with the ground truth masks, while FPs and FNs are significantly lower than in the other baselines.

D. Comparison of Model Parameters and Complexity

We evaluated the suggested method's computational efficiency by comparing its parameter count and floating-point operations (FLOPs) with the most advanced norms. Real-time applicability and training scalability are directly impacted by these metrics, especially in high-resolution satellite imagery

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TABLE VI
RESULTS OF AN ABLATION STUDY ON THE LEVIR-CD+ DATASET. BEST RESULTS ARE IN RED

Model	HSE	CRD	Precision	Recall	F1-score (%)	mIoU
Base (DeeplabV3+)	✗	✗	91.36%	88.89%	90.11%	82.78%
ARN	✗	✗	91.85%	89.94%	90.88%	84.16%
	✓	✗	92.14%	90.72%	91.42%	85.63%
ARN+ASPP	✓	✗	92.39%	91.68%	92.03%	87.92%
	✓	✓	92.58%	92.91%	92.74%	89.36%
ReSCD-Net(Ours)	✓	✓	**92.76%**	**93.36%**	**93.06%**	**90.83%**

where patch-based processing is prevalent. The complexity metrics of baseline methods shown in Table V and Fig. 8 are adopted from the original Edge-CVT publication [1], while the performance results of the proposed method are obtained from our experiments, early CNN-based Siamese methods, such as FC-Siam-Conc and FC-Siam-Diff, have very low complexity (<2 M parameters and <50 GFLOPs), but weaker detection performance in dense urban environments is the result of limited representational capacity. By increasing parameter depth, techniques such as DSAM-Net and BIT-Net increase recall and robustness while remaining lightweight in comparison to transformer-based models. Transformer-driven architectures such as EGCT-Net, ICIF-Net, and ChangeFormer significantly increase both parameters and FLOPs; ChangeFormer's global self-attention operations allow it to surpass 1600 GFLOPs. By combining CNN and transformer blocks, Edge-CVT lowers FLOPs in comparison to ChangeFormer, but it still has a high computational cost. In contrast, our suggested strategy shows an improved accuracy-efficiency tradeoff, achieving a 91.07% $F1$-score on EGY-BCD with just 120.6 GFLOPs and 35.2M parameters. While the CRD guarantees high boundary fidelity without the need for edge-specific modules, the learnable resizer minimizes redundant computation across scales. The suggested framework maintains competitive detection accuracy while requiring 7.3× fewer FLOPs than ChangeFormer and 2.5× fewer FLOPs than EGCT-Net when compared to the heavy baseline.

E. Ablation Study

We performed a thorough ablation study on the LEVIR-CD+ dataset to assess the effectiveness of each component in the proposed ReSCD-Net. Table VI summarizes the results, detailing how each configuration gradually incorporates a key module to assess its individual impact on change detection performance. Starting with the DeepLabV3+ baseline, the model attains an $F1$-score of 90.11% and an mIoU of 82.78%. The introduction of the ARN results in an $F1$-score of 90.88% (+0.77%) and mIoU of 84.16%, highlighting the advantages of resolution standardization in mitigating scale-induced pseudochanges. The integration of the HSE results in performance improvements, achieving an $F1$ score of 91.42% and mIoU of 85.63%. This suggests enhanced hierarchical feature representation and improved structural awareness. The incorporation of the ASPP results in a significant performance improvement, increasing the $F1$-score to 92.03% and the mIoU to 87.92%. This underscores the critical role of multiscale contextual reasoning in identifying subtle alterations in building structures. The integration of the CRD enhances boundary localization, achieving an $F1$ score of 92.74% and an mIoU of 89.36%. The results indicate that each module contributes positively and cumulatively, thereby validating the effectiveness of explicitly separating resolution normalization from semantic change modeling and integrating it with multiscale contextual and boundary-aware refinement.

VI. CONCLUSION

This study introduced ReSCD-Net, a resolution-aware deep learning framework that addresses two persistent challenges in BCD from high-resolution satellite imagery: scale inconsistency and insufficient contextual reasoning. The model combines an ARN for content-aware resolution standardization with a RASC based on a DeepLabV3+ backbone supplemented with ASPP. The integration allows ReSCD-Net to maintain spatial accuracy, record multiscale contextual linkages, and accomplish strong change localization across a variety of contexts. Extensive trials on three benchmark datasets, EGY-BCD, LEVIR-CD+, and DSIFN-CD, showed the model's applicability and adaptability to different spatial resolutions, geographic distributions, and structural complexities. ReSCD-Net outperformed CNN- and Transformer-based baselines such as ChangeFormer, EGCT-Net, and Edge-CVT, achieving 91.09% $F1$-score and 91.12% mIoU on EGY-BCD, 93.06% $F1$-score and 90.83% mIoU on LEVIR-CD+, and 93.32% $F1$-score and 87.48% IoU on DSIFN-CD. These findings demonstrate that resolution-agnostic preprocessing (ARN) and multiscale contextual reasoning (RASC with ASPP) significantly improve accuracy and robustness in complicated urban environments. The system will be enhanced by multimodal fusion of optical and SAR data to improve robustness under cloud cover and sensor variability, as well as semisupervised learning to enable scale deployment in data-constrained regions. Overall, ReSCD-Net is a reliable, scalable, and flexible technique for high-resolution BCD, filling the gap between CNN-based accuracy and Transformer-level context modeling in current remote sensing applications.

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The datasets and codes used in this article are available upon request from the authors.

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Mohamed Mosaad Elisy received the bachelor's degree in computer science (software engineering) from Wuhan University, Wuhan, China, in 2023, where he is currently pursuing the M.S. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS).

His research interests include computer vision and change detection using deep learning techniques.



Xiongwu Xiao (Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2018.

From 2016 to 2017, he worked as a Joint Ph.D. Student of applied mathematics at the University of California, Irvine, Irvine, CA, USA. In 2018, he joined LIESMARS, Wuhan University, as an Assistant Research Fellow, where he has been working as a Tenured Associate Professor since March 2025. He designed the first real-time photogrammetry system, named DirectMap, for unmanned aerial vehicle (UAV) image real-time online triangulation processing without ground control points (GCPs), and the real-time dynamic networking and real-time generation of mesh/digital elevation model (DEM)/digital orthophoto map (DOM) in April 2019. He owns 12 authorized invention patents and has published over 60 refereed articles. His research interests include photogrammetry and computer vision, real-time surveying and mapping, 3-D reconstruction, vector 3-D mapping of buildings, and deep learning.

Dr. Xiao was selected for the Chutian Scholar Program of Hubei Province in 2021 and the Youth Yangtze River Scholar Program in 2025. He has won the Second Prize of China Surveying and Mapping Science and Technology Award in 2019, the Gold Medal of China International "Internet Plus" Competition in 2022, and the Special Prize of China Geographic Information Technology Progress Award in 2023. He currently serves as an Associate Editor of the *Journal of Environmental Management* (JEM) and a committee member and the deputy secretary-general for several international and domestic academic organizations.



Mohamed H. Sharouda is currently pursuing the M.S. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China.

He is a Demonstrator with the Civil Engineering Department, Faculty of Engineering, Benha University, Benha, Egypt. His research interests include tropospheric delay modeling and GNSS meteorology applications using machine learning techniques.



Jianya Gong (Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from Wuhan Technical University of Surveying and Mapping, Wuhan, China, in 1992.

He is currently a Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan. He is an Academician with the Chinese Academy of Sciences, Beijing, China, in 2011. He was the President of the Commission VI of the International Society for Photogrammetry and Remote Sensing (ISPRS). His research interests include remote sensing image processing, spatial data infrastructure, and geospatial data sharing and interoperability.

Dr. Gong is currently serving as the Secretary-General for the Asia Geographic Information System Association (AGISA) and the Editor-in-Chief for the *Journal of Geodesy and Geoinformation Science*.



Deren Li received the Ph.D. degree from the University of Stuttgart, Stuttgart, Germany, in 1985, and the Honorary Doctorate degree from ETH Zurich, Zurich, Switzerland, in 2008.

He was the President of Wuhan Technical University of Surveying and Mapping, Wuhan, China, the Director of the Academic Committee, Wuhan University, Wuhan, and the Director of the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He is a Scientist of surveying, mapping, and remote sensing at Wuhan University. His research interests include geospatial informatics and spatiotemporal intelligence.

Prof. Li is an Honorary Member and a fellow of the International Society for Photogrammetry and Remote Sensing (ISPRS). He is a member of the International Eurasia Academy of Sciences and the International Academy of Astronautics. He enjoys dual memberships in both the Chinese Academy of Sciences and the Chinese Academy of Engineering. He has won the Highest Award "Brock Gold Medal Award" from the ISPRS society in 2022. At present, he is the Honorary Director of the Academic Committee, LIESMARS, the Director of the Collaborative Innovation Center of Geospatial Technology, an Honorary Chairperson of Wuhan Association for Science and Technology, and the Chief Scientist of the Optics Valley of China in Wuhan.

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